STATISTICAL LEARNING & DATA ANALYSIS

Comparative Study: Beta vs. Gamma Distributions

Name: Ammar Gharaf - D03000248, Amr Khalil D03000225, Aya Soundous Hechaichi D03000252

Under supervision of: Prof, Roberta Siciliano & Prof, EMILIANO DEL GOBBO

Università degli Studi di Napoli Federico II

The Gaussian (Normal) Distribution

The Gaussian (Normal) Distribution is the most important probability distribution in statistics, describing how values cluster around an average in a characteristic bell-shaped curve. Its prevalence in natural phenomena and statistical theory makes it a fundamental concept in data analysis.

1

Key Characteristics

- **Symmetry:** Perfectly symmetric around μ
- **Mean = Median = Mode:** All three coincide at the center
- **Unbounded:**
  - Defined on $(-\infty, +\infty)$
  - Can take negative or positive values
- **Bell-shaped:**
  - Highest density at the mean
  - Tails decrease smoothly

2

Mathematical Definition

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ: represents the center of the distribution (mean)
- σ: Standard deviation
- σ^2: represents the spread or variability of the distribution (variance)

Central Limit Theorem: This theorem states that the sum or average of many independent random variables will tend to be normally distributed, regardless of the original distribution of the variables themselves. This makes the Normal distribution a powerful tool for statistical and modeling.

Purpose of the Comparison

Our goal is to see the core differences between these two probability distributions, understanding their mathematical structures, the types of random variables they model, and how their parameters influence their shape, mean, and variance. We'll also examine the alignment between theoretical predictions and simulation results, and their relationship with the Gaussian Normal distribution.

We chose the Beta and Gamma distributions because their support and shape naturally match the type of real world data we are modeling: **proportions for Beta and positive waiting times for Gamma.**

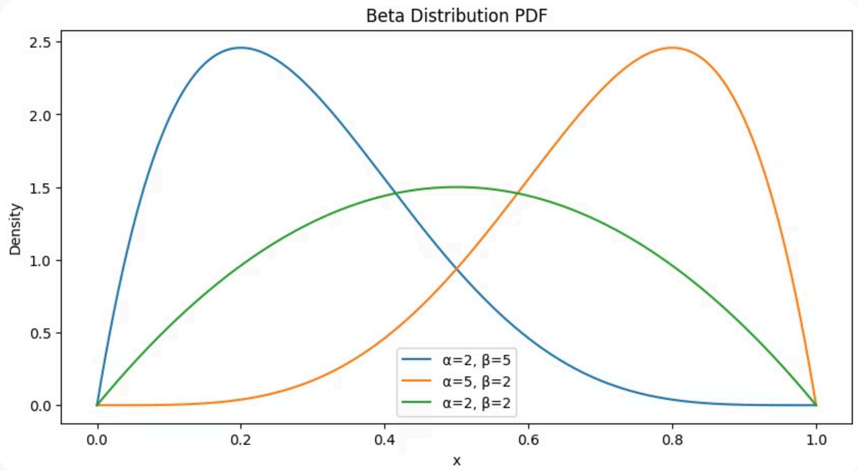
Both are continuous but the key difference is

Beta Distribution (Bounded - $0 \leqslant X \leqslant 1$)

Ideal for modeling proportions and probabilities, it's often bounded between 0 and 1.

Gamma Distribution (Unbounded - $Y > 0$)

Best suited for modeling positive waiting times or durations, often unbounded but positive.



Theoretical Definitions: Beta Distribution

The Beta distribution is defined by two positive shape parameters, ￿ and ￿, which exert significant control over its form within the interval [0, 1].

$$f_X(x;\alpha,\beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le x \le 1$$

Parameter Alpha

$$x^{\alpha-1}$$

Controls the distribution's behavior near 0.

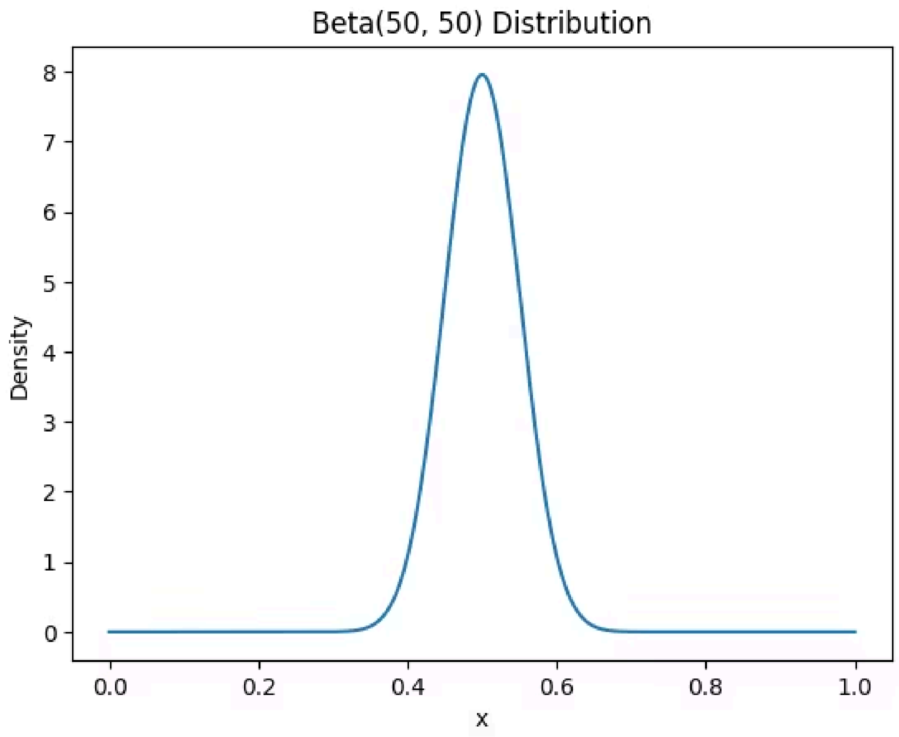
Parameter Beta

$$(1-x)^{\beta-1}$$

Influences the distribution's behavior near 1.

Symmetry

Occurs when alpha = Beta; skewness depends on their relative values.



When both shape parameters are large and equal, the Beta distribution becomes approximately symmetric and bell-shaped.

As shown in the Beta(50,50) plot, the distribution is sharply concentrated around its mean 0.5 and closely resembles a Gaussian distribution. This illustrates the connection between the Beta and Normal models.

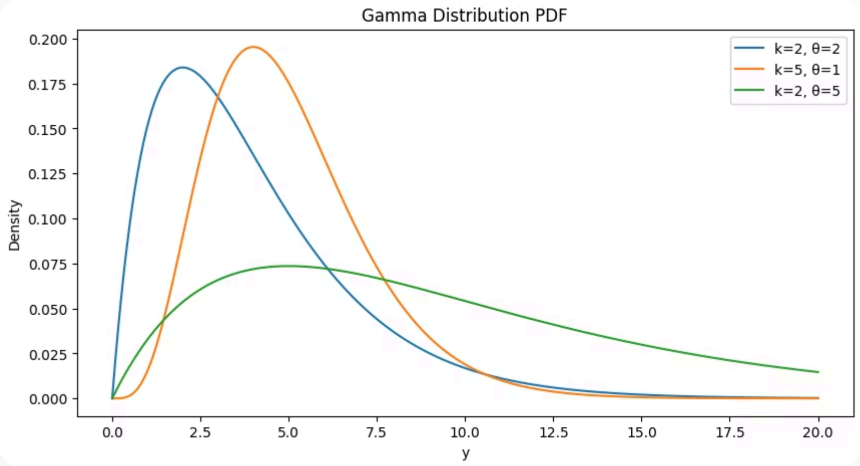
Theoretical Definitions: Gamma Distribution

The Gamma distribution is characterized by its shape parameter (k > 0) and scale parameter (θ > 0), crucial for modeling positive continuous variables like waiting times.

$$f_Y(y; k, \theta) = \frac{y^{k-1} e^{-y/\theta}}{\Gamma(k)\theta^k}, \quad y > 0$$

Shape Parameter (K): Governs the distribution's skewness and overall form.

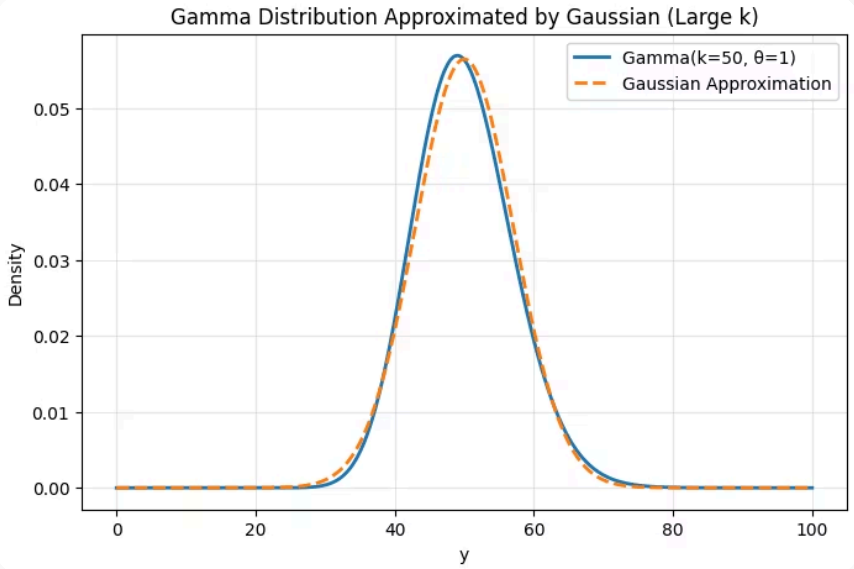
Scale Parameter (Theta): Determines the spread and scaling of the distribution.



Made with GAMMA

As shown in the plot,

According to CLT, for a Gamma random variable $Y \sim Gamma(k, \theta)$, the mean and variance are $E[Y] = k\theta$ and $Var(Y) = k\theta 2$. When the shape parameter k is large, the Gamma distribution can be approximated by a Gaussian distribution with the same mean and variance.



Beta vs Gamma

A crucial difference lies in the nature of the random variables each distribution is designed to model. This dictates their applicability to real-world phenomena.

Feature	Beta	Gamma
Support	[0,1] (bounded)	(0, ∞) (unbounded)
Variable type	Proportion (prob of success)	Time / duration
Shape	Flexible	Right-skewed
Interpretation	Probability, Conv rate	Waiting time
Mean	alpha / (alpha + Beta)	(alpha*Beta) / ((alpha + Beta)^2(alpha + Beta + 1))
Variance	K*theta	K*theta^2

Key Insight: Beta distribution's variance decreases as □ + □ increases, indicating greater certainty. Conversely, Gamma distribution's variance increases with its scale and shape parameters, reflecting higher dispersion.

Shape Behavior & Visual Interpretation

The inherent shape of these distributions strongly influences their suitability for different types of data.

Beta Distribution

Exhibits flexible shapes within the [0, 1] interval, adapting to left-skewed, right-skewed, or symmetric patterns. It's often used when values represent proportions or probabilities due to its bounded nature.

- Can be left-skewed
- Can be right-skewed
- Can be symmetric

Gamma Distribution

Always right-skewed for smaller □ values, making it ideal for modeling waiting times where shorter durations are more common. As □ increases, the distribution's shape becomes more symmetric.

- Always right-skewed (unless □ is large)
- Shape becomes more symmetric as □ increases

These distinct shape characteristics are vital for visually interpreting data patterns and selecting the most appropriate model.

Simulation Study: Empirical Validation

To validate our theoretical understanding, we conducted simulation studies for both distributions, generating 1000 random samples with various parameter values.

Generated Samples

1000 random samples from each distribution using different parameter sets.

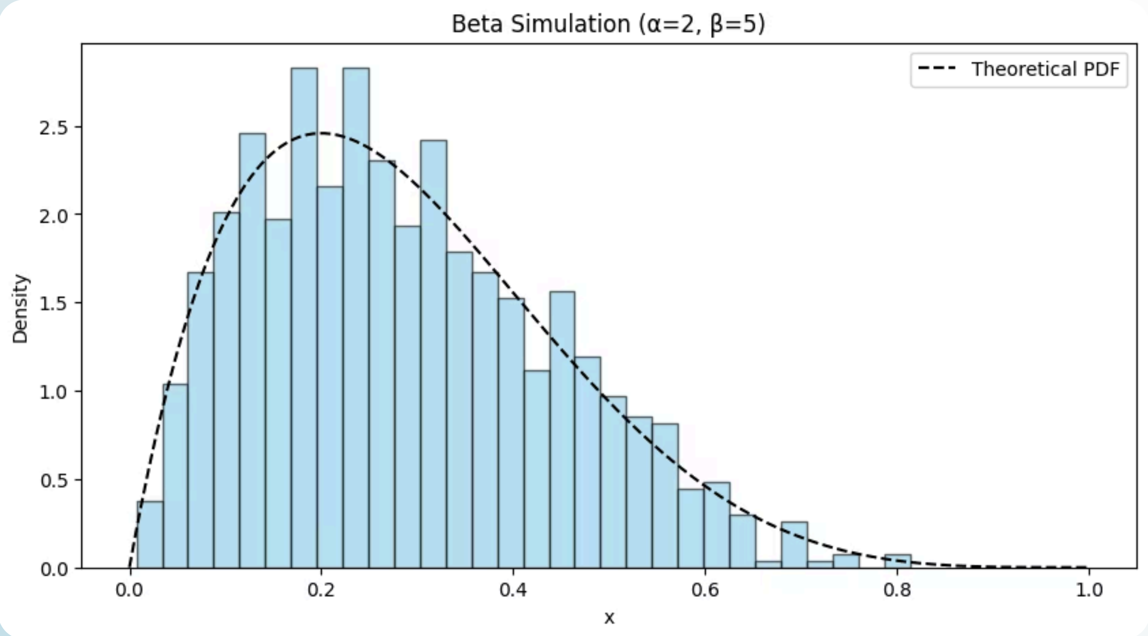
Visualized Data

Plotted histograms against theoretical PDFs for visual comparison.

Computed Statistics

Calculated sample mean and variance for comparison with theoretical values.

This empirical validation confirmed the Law of Large Numbers, showing that simulated means and variances closely matched theoretical predictions, reinforcing the consistency between theory and practice.



Beta Sample Mean: 0.29 - **Theoretical Mean:** 0.29 Beta & **Sample Variance:** 0.0241 - **Theoretical Var:** 0.0255

What the plot shows

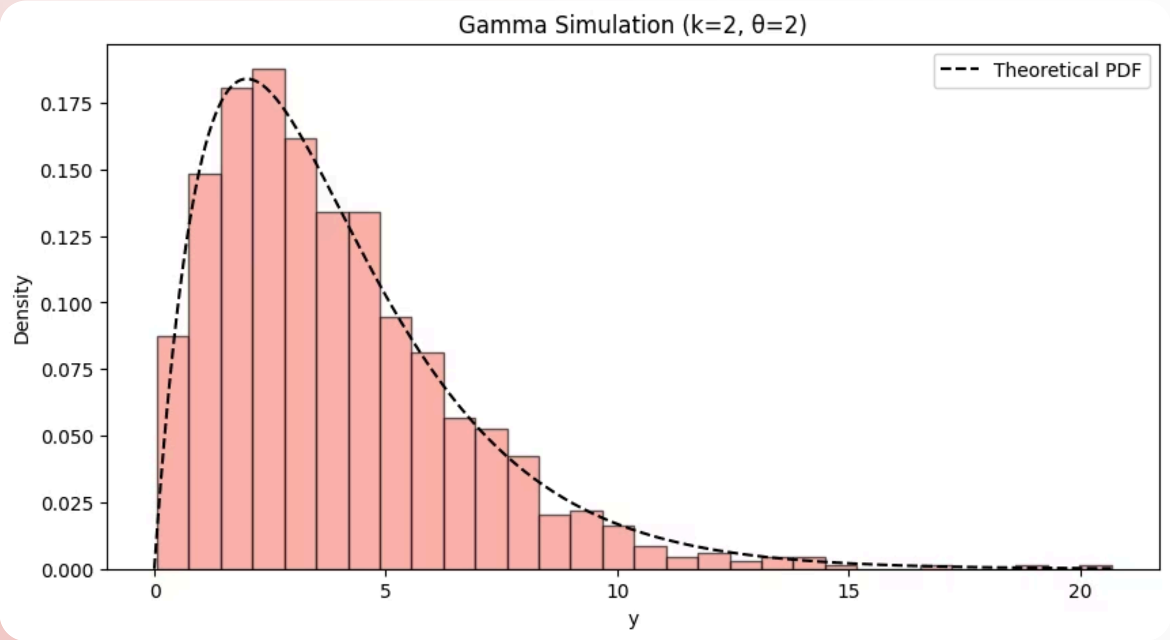
- **Histogram**: emp dist 1K simulated samples
- **Dashed curve:** theoretical Beta PDF
- **Strong overlap →** simulation matches theory

Shape and parameters

- Distribution is right-skewed
- Most values lie between 0.1 and 0.4
- $\beta > \alpha \Rightarrow$ more mass near 0
- Variable is bounded in [0, 1]

Statistical interpretation

- Emp dist approximates the theoretical model
- Small deviations due to random sampling
- Consistent with Beta distribution properties



Gamma Sample Mean: 3.92 - **Theoretical Mean:** 4.00 & **Gamma Sample Variance:** 7.33 - **Theoretical Var:** 8.00

What the plot shows

- **Histogram:** emp dist **1K simulated samples**
- **Dashed curve:** **theoretical Beta PDF**
- **Strong overlap → simulation matches theory**
- **K** controls **skewness & curve shape**
- **θ** controls **spread as high θ stretch the distribution**

Shape and parameters

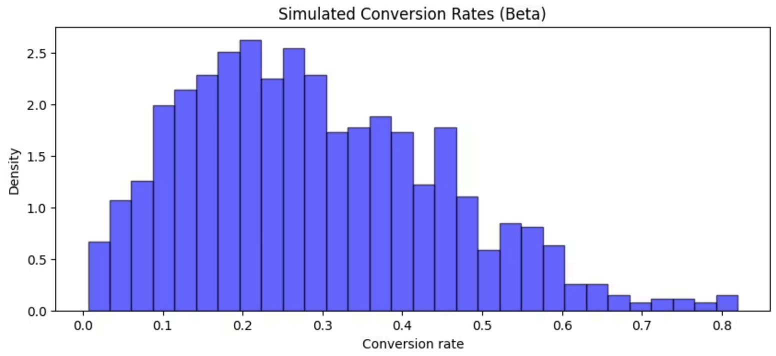
- Distribution is **right-skewed**
- Most values lie between **0 and 6**
- Long tail as large values are rare but possible
- Values are +ve & **unbounded.**

Statistical interpretation

- Sample mean & variance closely match theoretical values
- Confirms correctness of the Gamma model and simulation

Application Scenario & Model Choice

We analyze user behavior on an online advertising platform. Our goal is to understand two key random variables after a product advertisement is displayed:



Conversion Rate

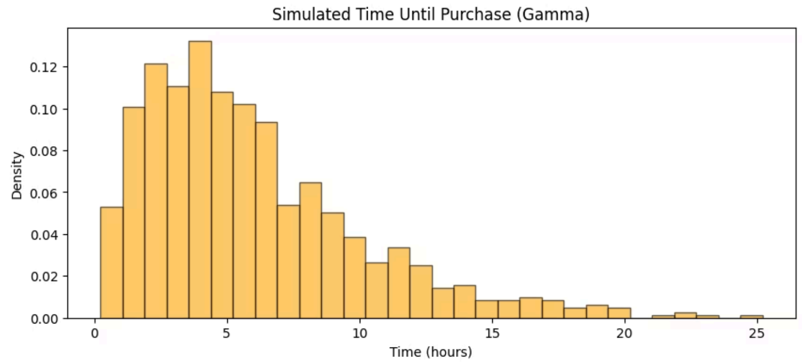
The proportion of users who click on the advertisement.

Modeled by: Beta Distribution

Parameters: alpha= 2, Beta= 5

- Variable is a proportion, bounded between 0 and 1.
- Parameters reflect that most users do not convert.
- Since $\beta > \alpha$, the distribution is **right-skewed**

> "Most users have low conversion probabilities, but a small fraction converts more frequently, which is typical in online advertising."



Time Until Purchase

The waiting time (H) between first click & the purchase.

Modeled by: Gamma Distribution

Parameters: k=2, Theta=3

- Variable is positive and continuous.
- Captures right-skewed waiting-time behavior, meaning most purchases happen soon.

> "Most users complete purchases shortly after clicking, while a smaller group delays their purchase significantly."

Connection to the Gaussian & Statistical Meaning

Both Beta and Gamma distributions, despite their distinct forms, exhibit a connections to the Gaussian (Normal) distribution under specific conditions.

Beta to Gaussian

Beta connected to the gaussian distribution as the two parameters are very big and equally.

Gamma to Gaussian

A Gamma variable with a sufficiently large shape parameter (□) approximates a Normal distribution. Also, Gamma is also the sum of exponential variables.

This comparative study underscores several critical statistical principles:

- Probability models must accurately reflect the underlying data structure.
- Distribution parameters are powerful tools that precisely control shape, spread, and skewness.
- The alignment of theory and simulation consistently reinforces statistical understanding.
- Different distributions serve unique statistical purposes, demanding thoughtful selection for robust analysis.

Final Conclusions

Key Takeaway

- Beta & Gamma follow the gaussian distribution under specific circumstances.
- Since Beta is bounded so it's used for prob/conv rate & Gamma is unbounded so it's used for timing/duration

Any Questions, Thanks.